

Sightline

Review Analyst • Conviction Integrity Unit

15 DAYS • 53 STOPS • GRADE 12 • SIGHTLINE

THE OPENING CARD

On a November night in 2019 Alma Cardoza watched a robbery from across Ferrier Street and later picked Elias Ward from six photographs. Her identification was the evidence that put Ward at the shop; no fingerprint, video or other physical evidence did. Police also recorded Ward agreeing with their account late in a long night of custody, and he took it back the next morning. He has served seven years of a fourteen-year sentence. You are the review analyst at the Conviction Integrity Unit. In fifteen days the board decides whether this identification can keep carrying the conviction. The file says Cardoza was twenty-two metres away under normal street lighting and later became absolutely certain. Your job is to separate what was measured, what was assumed, what changed after the robbery, and what the board is actually entitled to conclude. If the referral fails, Ward gets no new review until 2033.

What can the file actually support?

BEFORE THE DAY — GREET

A unit assembled for one case, and the review starts Monday

The integrity unit is a psychologist, an engineer, two lawyers and a retired detective who have never worked a case together. Ntuli, the deliberation study coordinator, wants you known to as many of them as you can before the file is opened, because every finding in this review depends on one of them accepting what another one measured.

PLAN CARD 121W

Monday. The review board sits in fifteen working days. Elias Ward has served seven years of a fourteen-year term for the robbery on Ferrier Street. One witness picked him out, and that identification is the only evidence that independently put him at the shop. The file also contains a late-night statement Ward retracted, but it was built from the police account rather than from a new fact only he supplied. Ines Baptiste wants the recommendation built on the procedure. Ren Okada wants the procedure reconstructed before anybody characterises it. Today you separate what the file says from what the file can actually support. By tonight, every load-bearing claim either has evidence behind it or is marked for the next fourteen days.

BRIEFING 21W

Fifteen working days. One identification put Ward at the robbery; before the board acts, determine what that identification can still carry.

WHAT YOU DECIDE 9W

Establish what the record asserts before anybody characterises it.

1.1 The signature and the thing it stands for

TESTING & RECORDS · DATA ROOM · A ROOM · ATTEST

SCENE 41W

Records custodian Nell Achterberg, the records custodian, opens the Ward file on the hall's long table. The cover says seven years served of fourteen. Every page is signed. The unit has fifteen working days, and this morning buys only three checks.

GUIDE 61W

Every page is signed and three checks is what the morning buys. Open each claim and read what stands behind it: an observation, a document, or another signature. Verify what the morning allows, and hold whatever the store cannot stand behind. Fifteen working days is the whole review, so what you hold now is what the rest of it works on.

ASK 13W

Verify what the morning allows, and hold whatever the store cannot stand behind.

BEHIND THE BUTTON 148W

What a records store can and cannot attest. It can say that a document exists, when it arrived and who signed it. It cannot say the thing the document asserts is true. Those two are recorded in the same handwriting, on the same page, and a file read quickly does not distinguish them.

Why the cover matters. 'Seven years served of fourteen' is the fact that makes this urgent, and it is also a claim on a cover sheet. Whether it is backed by the custodial record or copied from an earlier summary is exactly the kind of thing three checks are for.

Why holding is not doubting. A held claim is one the review will not build on until it is verified. It may well be true. What it cannot be is load-bearing, and marking that early is what stops fourteen days of work resting on it.

TAKES AS READ

a signature records that somebody accepted a claim at the time · a review has less time than a file has pages

VERDICT 77W

A file compresses months of work into signed claims. A signature shows who accepted a claim, not whether the condition was true. That difference matters most for claims carrying the case. Distance and lighting are both decisive and neither has a repeatable measurement behind it. The certainty statement is different: it is a real record, but its date changes what it can support. With only three checks, verification belongs where an unsupported claim could change the review.

TAKEAWAY 16W

A record is evidence that a claim was accepted, which is not evidence for the claim.

1.2 One in six is not an error rate

IDENTIFICATION PROCEDURE · IDENTIFICATION SUITE · A PERSON · BALLPARK

SCENE 40W

Marcus Ilori, the identification procedures adviser, has the 4 December 2019 array record on the table. Six photographs were laid out together. The witness was asked whether the man she saw was among them. It was standard procedure that year.

GUIDE 67W

Put a number on what the array asks. If the man she saw is not among the six and she picks anyway, the chance she lands on the suspect rather than a filler is one over the number of photographs. Pick the count and work it out. One tile is the number of fillers, which is what the suspect is being compared against rather than the divisor.

ASK 23W

If the man she saw was not among the six and she picked anyway, how often would the suspect be the one picked?

BEHIND THE BUTTON 138W

What simultaneous presentation can change. Seeing all six photographs together makes side-by-side comparison easy, but a witness can still reject the array. A may-not-be-present instruction matters because it makes rejection explicit rather than implying that one of the six must be the answer.

What one in six means. If the offender is absent, the witness nevertheless makes a choice, and all six photographs are equally plausible, the suspect receives one of six choices on average. That is a conditional equal-choice baseline, not a measured false-identification rate and not a floor that every simultaneous lineup must exceed.

What the fillers do. Fillers who do not fit the witness's prior description make the suspect stand out. That can make the effective size much smaller than six, which is why the later mock-witness test matters more than the printed array size.

TAKES AS READ

a lineup contains one suspect and a number of people known not to be the offender

VERDICT 131W

Six photographs shown together can encourage relative comparison: which face looks most like the memory. A witness can still reject the lineup, especially when explicitly told that the offender may be absent, which is why the missing caution matters. Put a number on it. If the man she saw is not among the six and she picks regardless, the chance she lands on the suspect rather than a filler is $1 \div 6 \approx 0.17$. That figure is only the equal-plausibility baseline. Fillers nobody would plausibly choose make the suspect stand out and can push the suspect-pick probability far above one in six. The protection therefore comes from fair fillers, a clear may-be-absent instruction, blind administration and a recorded immediate confidence statement — not from the printed number six by itself.

TAKEAWAY 22W

The printed size of a lineup is not a false-identification rate; fair fillers, instructions and administration determine how meaningful a choice is.

1.3 What a clean file would look like

INFLUENCE & THE ROOM · DELIBERATION ROOM · A ROOM · CHOICE

SCENE 39W

Grace Ntuli stops the first meeting four minutes in. Everybody in the room came here to find bad procedure, she says, and the file has not been opened yet. She wants something written on the whiteboard before it is.

GUIDE 56W

Four things writing the standard first might protect. Ask of each whether it is about the room or about the paperwork. Everybody here came to find bad procedure, and expectancy changes what people notice without making them invent anything. A standard written first turns a search into a comparison, and gives contrary evidence somewhere to land.

ASK 15W

What does writing down the features of a sound 2019 procedure protect the review from?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

people find more of what they are looking for than of what they are not

VERDICT 73W

Expectancy changes what people notice and remember without making them fabricate evidence. Reviewers who expect bad procedure naturally recognize faults faster than facts pointing the other way. Writing the standard first creates a comparison instead of a search. The same logic supports preregistration in experiments. Evidence that contradicts the room's expectation already has a place to land. That makes it harder to move the standard after the file starts telling a persuasive story.

TAKEAWAY 17W

Write down what would count against you before you look, or you will not recognise it later.

END OF THE DAY 14W

A file states things; the question is always which of them anything stands behind.

What could she see from twenty-two metres?

PLAN CARD 103W

Tuesday. The bay is finished. Dawit Bekele, the reconstruction technician, has built the Ferrier Street corner at full size. Kerb line, doorway and column, all of it. Halina Frey, the vision scientist, wants the file's own claim tested before anybody argues about it. The case summary says twenty-two metres. Nobody has ever put that number through anything. Today you work out what a face is at that range, measured as an angle at the eye. Then you read what Alma Cardoza's first account actually said. Ward is thirty-four. He has served seven years on what one person could make out across a street.

BRIEFING 22W

Start with the file's own twenty-two-metre claim and ask what visual information that distance could supply before memory or lineup procedure enters.

WHAT YOU DECIDE 12W

Turn the file's own distance into what an eye could have resolved.

2.1 Twenty-two metres, as an angle

PERCEPTION & THE SCENE · RECONSTRUCTION BAY · A PERSON · BALLPARK

SCENE 36W

Vision scientist Frey lights the reconstruction bay and runs a tape to the twenty-two-metre floor mark. She puts a face at the doorway and asks what its features become at an eye across the rebuilt street.

GUIDE 61W

Five numbers, and three of them belong to other questions. The width of a whole face. The height of the doorway. The viewing time. Ask of each whether this angle needs it. A face arrives at the eye as an angle, not as metres. So the distance in the file is not background. It sets what was available before memory enters.

ASK 16W

Estimate the angle the gap between a person's eyes subtends at the distance the file records.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

the eyes of an adult face are about six centimetres apart · an angle in radians is a length divided by a distance

VERDICT 77W

A face reaches the eye as angles, not metres. Angular size is the feature's size divided by its distance, so every step backward shrinks the usable detail. The eye gap here spans about three milliradians, or ten arc minutes. That can make a face visible as a face. Identity depends on smaller relationships inside that angle. The file's distance is therefore not background description. It sets the scale of the information available before memory enters the problem.

TAKEAWAY 17W

Doubling the distance halves the angle, and the angle is what the eye has to work with.

2.2 Six seconds is not six measurements

MEMORY & INTERVIEW · INTERVIEW SUITE A · A ROOM · ALLOCATE

SCENE 35W

Memory scientist Ren Okada opens Alma Cardoza's first full account at the doorway paragraph. Six seconds, by her estimate, separate the shout from the man crossing the kerb. The bay clock is set to six.

GUIDE 65W

Six seconds of attention, and a list of things that would take longer than that between them. Spend it, and read off what the account can then contain. Nothing on the list is a wrong thing to have looked at; what the six seconds cannot buy is the price of what they did. Fixating a face and reading a number plate are not simultaneous acts.

ASK 17W

Use the six-second teaching budget. What trade-offs does it make plausible, and what can it not prove?

BEHIND THE BUTTON 114W

Why this is a teaching budget. Eyes move several times each second, but a transcript cannot reconstruct where Cardoza looked on each movement. The board makes the competition visible without pretending those exact costs were measured in 2019.

Why the items compete. Reading a plate, inspecting a face and tracking a threat all draw on limited visual attention. A threatening object often attracts attention, but the size of that effect varies across people and events.

What the review is allowed to infer. Details that were not well attended are less likely to be encoded in retrievable form, but this model alone cannot say which ones those were. That is why the later chronology matters.

TAKES AS READ

memory is built from what was attended to during an event

VERDICT 93W

Attention is limited and encoding depends on where it goes. The allocation board is an intentionally simplified teaching model, not a measured fixation ledger for Cardoza. A visible threat can compete strongly for attention, while a distant face, a number plate and the surroundings each require their own looks. The useful conclusion is therefore a vulnerability, not a verdict: detailed memory for all of them should not be assumed equally strong after a six-second event. Later documents must be dated source by source to learn whether any particular detail entered after the robbery.

TAKEAWAY 19W

A short, threatening event predicts competition for attention; it does not identify which later detail came from which source.

2.3 Predict it before the forty cards turn

IDENTIFICATION PROCEDURE · IDENTIFICATION SUITE · A ROOM · VERIFY

SCENE 31W

Marcus Ilori lays Cardoza's first description beside the six 2019 photographs at the identification bench. Forty mock-witness cards are face down beside them. Nobody holding a card has seen the robbery.

GUIDE 55W

Commit to the share of mock witnesses you expect to pick the suspect from the description alone, before any card is turned. It cannot be changed afterwards. Then turn the forty cards and read the measured distribution. Nobody holding a card saw the robbery, so what comes back measures the array rather than anybody's memory.

ASK 20W

What suspect-pick share do you expect from the description alone, and does the measured distribution support a fair six-person array?

BEHIND THE BUTTON 136W

What a fair array looks like in numbers. Six photographs, one of them the suspect, and a description that does not single anybody out should send about one witness in six to each. A share much higher than that means the array itself is pointing, and the description is doing the work rather than the memory.

Why mock witnesses who saw nothing are the right test. They cannot recognise anybody, so anything above chance comes from the procedure: fillers who do not match the description, a photograph that stands out, an instruction that implies the offender is present.

Why the prediction is locked first. A distribution read and then explained is a distribution that can be made to support whatever the reader already believed. Committing first is what makes the forty cards evidence about the array.

TAKES AS READ

a lineup is built to be consistent with the description already given

VERDICT 74W

A fair lineup should not point to the suspect through the description alone. Mock witnesses are useful because they never saw the event. They can respond only to the description and the photographs. If one photograph attracts most of them, the array is doing part of the identifying. Cardoza's description is broad, so visual matching alone cannot establish fairness. Measuring where mock witnesses land turns that concern into an empirical property of the procedure.

TAKEAWAY 24W

A procedure can be tested by people who know the description but not the event; where they land measures the array rather than memory.

END OF THE DAY 19W

A face at a distance arrives as an angle, and the angle is small long before the person is.

Where does this test fall to chance?

PLAN CARD 83W

Wednesday. Twelve volunteers have worked the reconstruction at seven marked distances. They could describe something from every mark, but description and identification are not the same task. Today you find the first station where this small sample no longer beats the six-way chance baseline. Then you learn what dim adaptation trades away and why “I saw his face” is several claims bundled into one. The result will be a bound the review can use, not a magic distance at which human recognition stops.

BRIEFING 26W

The bay can show where this group’s performance begins to look like guessing. It cannot draw a universal line beyond which nobody can recognise a face.

WHAT YOU DECIDE 18W

Use a small reconstruction study to bound identification performance without turning one sample into a universal human cutoff.

3.1 Where this sample stops beating chance

PERCEPTION & THE SCENE · RECONSTRUCTION BAY · A ROOM · PROBE

SCENE 61W

Twelve volunteers stood at each of seven marks down the bay floor and described a face at the far end. Later each tried to pick that person out of six photographs. Describing and identifying are not the same task. Picking one of six by chance alone succeeds about twice in twelve, and Bekele wants the distance where naming stops beating that.

GUIDE 62W

Nothing is read for you. Press Read on any mark and it reports what the twelve described from there, and how many of them later picked the right face out of six. Take at least three. Judge the naming rate against what chance alone would give rather than against zero. Then click the first mark where naming has stopped working, and commit.

ASK 16W

At which station does this small sample first perform no better than the six-way chance expectation?

BEHIND THE BUTTON 198W

Why the two tasks come apart. Build, clothing, gait and hair are separate features, and any one of them survives a long look at low detail. Recognising a particular face depends on the relations between its parts — the spacing of the eyes against the width of the nose — and those relations blur first. So a witness can be entirely accurate about a jacket at a distance where the face was never available at all.

What chance looks like in this data. Twelve people picking blindly from six photographs will average two correct, and will sometimes give three or one. A station scoring two out of twelve is therefore reporting no identification at all, and one scoring three is reporting almost none. Zero is not the benchmark, and treating it as one makes a witness sound better the further away they stood.

Why the distance is worth writing down. Testimony arrives as a description and an identification in one sentence, and a jury hears both as one claim. Knowing the distance at which naming failed in a controlled test on this same bay gives the court a number to put against the sentence rather than an impression.

TAKES AS READ

features can be described without the person being identifiable · the same face was used at every distance

VERDICT 78W

Large features such as build, clothing and gait survive distance longer than identity-carrying facial relationships. In this authored reconstruction, correct naming falls to two of twelve at twenty metres, while broad description survives. Because the sample is small, the scientifically defensible conclusion is that identification performance is no longer clearly above chance under these test conditions. The study

does not establish a hard human recognition distance, and it does not by itself tell what Cardoza saw in 2019.

TAKEAWAY 18W

A controlled reconstruction can bound performance under its own conditions; it does not create a universal distance limit.

3.2 What the dark takes

STRESS, SLEEP & THE BODY · PHYSIOLOGY BAY · A ROOM · CHOICE

SCENE 39W

Priya Raghunathan, the psychophysicologist, shuts the booth door and waits ten minutes. She can then find the chart on the far wall but cannot name its colours. She wanted the room to experience that loss before reading about it.

GUIDE 58W

Four things a dark scene might take away. Ask of each whether it fits what the two receptor types do. Cones give colour and fine detail and need light. Rods pool signals, so they buy reach and lose resolution. That means a dark street does not simply dim a face. It removes some things and leaves others intact.

ASK 17W

What is most strongly degraded as vision shifts from cone-dominated toward rod-dominated vision in very dim light?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

the retina has two kinds of light-sensing cell with different jobs · illuminance, and the light a scene actually had — taken as read

VERDICT 76W

Cones provide colour and fine central detail, but they need more light. Rods work when illumination falls and trade detail for the ability to see at all. They also pool signals across retinal areas, which makes the result coarser. A dark street therefore does not simply dim a face. It removes colour and fine relationships while preserving movement, outline and broad form. That pattern can support confident description of some features while leaving identification detail weak.

TAKEAWAY 20W

Sensitivity in a dark place is bought, and the price is paid in what the eye can no longer separate.

3.3 One claim, or four

INFLUENCE & THE ROOM · DELIBERATION ROOM · A PERSON · CHOICE

SCENE 37W

Grace Ntuli plays a panel recording in the hall. A volunteer juror repeats 'she saw his face' eleven times across forty minutes. By the end, nobody in the room is asking how far away the face was.

GUIDE 59W

Four accounts of why one phrase is hard to argue with. Ask of each whether it is about the words or about who said them. The sentence contains a distance, a light level, a duration and a recognition. Three of those can be checked without asking the witness anything. Bundled together, none of them can be examined at all.

ASK 16W

Why does the phrase "she saw his face" make a room's decision harder rather than easier?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a repeated phrase becomes the thing a room is deciding about

VERDICT 72W

'She saw his face' bundles several claims into one sentence. It contains distance, illumination, viewing time and recognition. Treating the phrase as one fact forces a reviewer to accept or reject all four together. Separating them is not an attack on honesty. It is how each claim becomes testable. Distance, lighting and timing can be checked without asking the witness again. Recognition is the part that remains tied to memory and procedure.

TAKEAWAY 19W

"I saw his face" bundles a distance, a light level, a duration and a recognition into one unarguable sentence.

3.4 Hold a dim-light test steady

PERCEPTION & THE SCENE · RECONSTRUCTION BAY · A ROOM · HOLD

SCENE 45W

Before anyone claims to have reconstructed 2019, the bay team runs a control at five lux. Lamps warm and drift, somebody opens the corridor door, and the dimmer is the only control. This is a calibration drill, not yet a claim about the robbery night.

GUIDE 36W

Hold the bay at the five-lux test condition, inside the band on the meter. The point is to prove the reconstruction can hold a stated condition before the team decides what historical condition belongs in it.

ASK 13W

Hold the bay at the five-lux test condition while the walk is done.

BEHIND THE BUTTON 113W

Why this has to be held rather than set. Lamp output and stray corridor light change during a run. A reconstruction that drifts cannot tell whether the outcome changed because of the participant or because the room changed.

Why the level is labelled a test condition. Five lux is deliberately dim and useful for showing how visual performance changes, but the historical light level on 14 November has not yet been established. A stable model can still be a model of the wrong night.

Why the door is a disturbance. Stray light adds to the scene and cannot be subtracted from a participant's response afterwards. The meter is what makes the run repeatable.

TAKES AS READ

a scene can be reproduced at a measured light level

VERDICT 71W

The value of a reconstruction begins with control. The bay first has to show that it can hold a stated light level while lamps warm and doors change the stray light. Five lux is a deliberately chosen dim test condition, not the claimed 2019 illuminance. That distinction matters: repeatability answers whether the experiment is controlled; the maintenance record and scene evidence answer whether the chosen condition belongs to the historical night.

TAKEAWAY 20W

A reconstruction can be perfectly controlled and still model the wrong historical condition; control and historical validity are separate questions.

There is a distance at which a face becomes a shape, and it is closer than anybody expects.

If the lamp was working

BEFORE THE DAY — FOLLOW

Walk the route the witness walked, at her pace

Cardoza came out of the shop, crossed to the corner and stopped. The reconstruction is only worth what the route is worth. Stay with whoever is walking it and stay behind them — somebody walking in front of the sightline is standing exactly where the argument is.

PLAN CARD 98W

Thursday. Baptiste argues that the unit already has evidence that does not depend on asking Cardoza anything new. She is right about the method and not yet about the answer. Today you calculate the doorway illumination implied by the utility specification if column 4471 was working, then separate that model from the unmeasured word “normal.” You also ask what threat can plausibly do to attention without turning a population tendency into a diagnosis of one witness. By tonight the light claim becomes a testable assumption waiting on a historical record, not a fact copied from the scene report.

BRIEFING 25W

The scene report says the lighting was “normal.” Today you calculate what that would mean if the listed streetlamp was working—and leave the premise open.

WHAT YOU DECIDE 20W

Model the light implied by the file without mistaking a correct calculation for proof that the lamp was actually on.

4.1 How much light the corner had

PERCEPTION & THE SCENE · RECONSTRUCTION BAY · A ROOM · BALLPARK

SCENE 29W

Frey has the column's specification from the utility: a 1,800 candela lantern, twelve metres from the shop doorway along the kerb. The scene report's word for this is "normal".

GUIDE 58W

Five numbers, and three of them are the witness's distance and two figures offered for comparison. Ask of each whether this calculation needs it. The distance that matters runs from the lamp to what it lights, not from the eye to the scene. And note what the answer is not: it assumes the lantern was on that night.

ASK 13W

Estimate the illuminance the column put on the doorway when it was working.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

light from a single source spreads over an area that grows with distance

VERDICT 74W

Light from a source spreads over increasing area, so illuminance falls with the square of distance. The relevant distance runs from the lantern to the doorway it lights. Cardoza's distance belongs in the angular-size calculation instead. A working 1,800-candela lantern at twelve metres gives 12.5 lux. That is dim street lighting, not darkness. The calculation is sound. Whether the lantern was actually on that night is a separate factual question the calculation cannot answer.

TAKEAWAY 21W

Illuminance is set by the distance from the lamp to the thing lit, not by the distance to whoever is looking.

4.2 The middle of the event and its edges

STRESS, SLEEP & THE BODY · PHYSIOLOGY BAY · A PERSON · DIAGNOSIS

SCENE 35W

Priya Raghunathan has four parts of Cardoza's account on the screen. One shows detailed recall, one shows nothing, one is wrong, and one was never mentioned. She wants one mechanism that fits the whole pattern.

GUIDE 57W

Four readings, and one of them is an absence. Nothing at all about the second man. Ask of each candidate whether it explains dense detail in one place and nothing in another. Two options say the account is bad throughout, which the stable broad features contradict. The pattern is structured, and a structured pattern has a mechanism.

ASK 11W

Which explanation fits all four readings rather than the loudest one?

BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

TAKES AS READ

a threat triggers a fast alarm response before any decision is made • attention narrows under threat

VERDICT 83W

Threat can draw attention toward the most urgent object and reduce processing of other information. That makes weapon-focus-like trade-offs a plausible explanation for a record rich in central threat details and weaker elsewhere. But it is a tendency, not a rule that tells the review exactly what Cardoza encoded. The point of the pattern is to generate a testable vulnerability: peripheral and facial details deserve more caution, while later chronology is still needed to determine whether any particular detail changed after the event.

TAKEAWAY 24W

Threat can narrow attention and make some details more vulnerable; it does not tell the reviewer exactly which details were or were not encoded.

4.3 A word where a reading should be

TESTING & RECORDS · DATA ROOM · A ROOM · CHOICE

SCENE 44W

Unit Director Ines Baptiste stands on the rebuilt corner beside the lux meter and the scene report. The report says only 'street lighting normal.' No night reading survives. She asks how much longer Ward should serve while the unit treats that word as data.

GUIDE 59W

Four objections to one word. Ask of each whether it questions the officer or the record. Normal is a judgement against something unstated. It could mean the lamps looked usual, or that nothing seemed odd, or that the road was familiar. Those imply different physical conditions, and none of them gives a number anybody can go back and check.

ASK 15W

Why can the review not simply accept "street lighting normal" as evidence about the light?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a measurement can be repeated by somebody else and a description cannot

VERDICT 73W

'Normal' is a judgment against an unstated comparison. It could mean the lamps looked usual, that nothing seemed unusual, or simply that the road was familiar. Those meanings imply different physical conditions. None gives a lux value another reviewer can reproduce. A meter reading carries a defined quantity and unit. Seven years later, somebody else can make the same measurement. A descriptive word cannot supply that missing standard, however sincere the writer was.

TAKEAWAY 22W

A word describing a condition and a reading of that condition are different kinds of evidence, and only one can be checked.

4.4 What a clean file would look like — Review

INFLUENCE & THE ROOM · DELIBERATION ROOM · A ROOM · CALLBACK · CHOICE

SCENE 33W

Ntuli reads back the whiteboard from the first morning. The room agreed then what a sound 2019 procedure would have looked like, and she wants to know what the list is for now.

GUIDE 60W

Four things a written standard buys. Ask of each whether it helps the room or helps a reader who distrusts the room. A comparison against a written standard can be argued with in two places: the standard, or whether the file meets it. A judgement cannot be argued with at all, which is the same protection a prespecified analysis gives.

ASK 15W

What does the review gain by comparing the file against a standard it wrote first?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

a standard agreed in advance can be compared against afterwards

VERDICT 81W

A finding stated as a comparison against a written standard is checkable by somebody who thinks the reviewers are biased: they can dispute the standard, or dispute whether the file meets it, and both arguments are out in the open. A finding stated as a judgement is not checkable at all. The protection is the same one a prespecified analysis gives an experiment, and it works for the same reason — the standard cannot be adjusted after the answer is known.

TAKEAWAY 15W

A standard written before the reading turns the report into a comparison anybody can repeat.

END OF THE DAY 15W

Most of what decides an identification can be measured without ever asking the witness again.

What entered the first account?

PLAN CARD 96W

Friday. Cardoza's first full interview happened four days after the robbery. Ren Okada has the recording and a transcript nobody has checked against it. Yara Solomon has found something else: Cardoza's husband sat in for the whole hour. Today you rebuild an interview that protects a witness's own account, test how agreement in a listening room can alter later repetition, and read the instruction Cardoza received before the photographs appeared. The goal is not to prove that any one later detail was contaminated. It is to identify which pathways were open before the lineup ever began.

BRIEFING 12W

The account everybody has argued about since was recorded on 18 November.

WHAT YOU DECIDE 14W

Establish how the first full account was taken, and who was in the room.

5.1 The order an account is taken in

MEMORY & INTERVIEW · INTERVIEW SUITE A · A ROOM · SEQUENCE

SCENE 39W

Ren Okada lays five interview steps along the corridor floor and queues the 18 November recording. In the audio, the second question arrives eleven seconds after Cardoza begins. He asks the room to rebuild the order before replaying it.

GUIDE 57W

All five steps will happen, so ask what each one protects from the ones after it. Once somebody else's words are in the room, the account cannot be returned to what it was. So ask of each step whether it adds information or only invites it. And one of them has to happen before any photograph exists.

ASK 13W

Order the steps of an interview that protects what the witness actually has.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a question can introduce a detail that was not in the witness's own account · how a lineup is built: fillers, and the ones nobody would pick — taken as read

VERDICT 74W

Interview order protects later information from earlier contamination. First, make 'I do not know' an acceptable answer. Then take uninterrupted free recall before adding anybody else's words. Open questions can follow because they expand material the witness already raised. Closed questions come later and only on existing topics. Confidence must be recorded before photographs or feedback arrive. Once outside information enters, the original state of the account cannot be recreated by asking more carefully.

TAKEAWAY 18W

Every question narrows what comes next, so the questions go last and the witness's own account goes first.

5.2 Eleven small agreements

INFLUENCE & THE ROOM · DELIBERATION ROOM · A ROOM · CONTROL

SCENE 36W

Grace Ntuli plays the 18 November interview in the listening room. Cardoza's husband is present throughout and agrees at eleven points. A second console can replay the same account with one feature of the room changed.

GUIDE 64W

The number you are watching is how many details from the first account survive into the second one. Three features of the listening room can be changed, one at a time, and the same recording is replayed each time. A response counts only if it is larger than the run-to-run scatter. Change one, replay, restore it, and name the feature the surviving details follow.

ASK 19W

Which room feature is causing reinforced details to survive, and does the effect return when that feature is restored?

TAKES AS READ

people adjust what they say to the reaction they are getting · how a lineup is built: fillers, and the ones nobody would pick — taken as read

VERDICT 80W

Social feedback can shape what a speaker repeats without deliberate coaching. In this controlled volunteer run, changing listener agreement while holding the interviewer and timing fixed changes which details survive into the second telling, and reversing the cue reverses the effect. That is causal evidence for the mechanism in the rig. The 2019 interview was not a controlled experiment, so the review may say that an influence pathway existed—not that it can reconstruct which sentence, if any, the husband changed.

TAKEAWAY 25W

A controlled study can establish that social feedback can move an account; the case record still has to show whether that mechanism reached this witness.

5.3 The man may not be here

IDENTIFICATION PROCEDURE · IDENTIFICATION SUITE · A PERSON · PROTOCOL

SCENE 38W

Ilori has the 2019 array form. There is a line for the witness's signature and no line for the caution that the person she saw might not be among the photographs. In 2019 there was no such line.

GUIDE 57W

Match each outcome an array can produce to what it actually costs, and to who bears the cost. Two of the four are errors and two are correct answers, and the caution is what changes the rate of one specific error without changing the others. Sorting them is how the missing line stops being a paperwork point.

ASK 12W

Match each array outcome to what it costs and who carries it.

BEHIND THE BUTTON 157W

Why an array has four outcomes and not two. The offender is either in the array or not, and the witness either picks the suspect, picks a filler, or declines. That gives a hit, a miss, a filler pick and a correct rejection — and the four are not equally visible afterwards, which is why a procedure can look successful while producing the wrong ones.

What the caution changes. A witness can reject an array even without the instruction, but explicitly saying that the offender may be absent reduces the pressure to treat one of the photographs as the required answer. That matters most when the offender is absent.

Why a filler pick is the useful signal. It is the one error the system can see: nobody charges a filler, so a filler pick tells the investigation the witness was guessing. An identification of the suspect under the same guessing produces a charge and looks like evidence.

TAKES AS READ

a witness expects an investigation to have found somebody before showing an array · how a lineup is built: fillers, and the ones nobody would pick — taken as read

VERDICT 109W

An array can produce a hit, a miss, a filler pick, a suspect pick when the offender is absent, or a correct rejection. Those outcomes are not equally visible later. A filler pick is obviously an error because the filler is known innocent; a suspect pick can travel through the case as evidence. The missing caution matters because a clear may-not-be-present instruction legitimises saying “none of them” and can reduce choosing pressure. It does not eliminate every error, and its absence does not force a witness to choose. The defensible finding is that a protective instruction was missing from a procedure already vulnerable to a suspect who stood out.

TAKEAWAY 19W

A may-not-be-present instruction makes rejection explicit and reduces pressure to choose; its absence does not make a choice inevitable.

5.4 What the dark takes — Review

STRESS, SLEEP & THE BODY · PHYSIOLOGY BAY · A ROOM · CALLBACK · CHOICE

SCENE 32W

Raghunathan asks the reverse question. Two lux takes colour and fine detail away; the room has spent a fortnight on what is lost, and she wants the other half of the list.

GUIDE 61W

Four things a witness at two lux might get right, and the room has spent a fortnight on what is lost. Ask of each whether rods can carry it. They pool signals, which buys reach and costs resolution, and they are missing from the centre of gaze. So the useful question is which part of the account the conditions could support.

ASK 13W

What is a witness at two lux most likely to be right about?

BEHIND THE BUTTON 195W

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Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

rods are sensitive, coarse and colour-blind

VERDICT 85W

Rods pool their signals, which buys reach in the dark and costs resolution, and they are absent from the very centre of gaze — so a dark scene delivers movement, outline and heading well and everything fine badly. A report that treats a witness as reliable or unreliable as a whole gets this wrong in both directions. The useful question is always which part of the account the conditions could support, and gait and direction are the parts that survive two lux at thirty-four metres.

TAKEAWAY 19W

A dark-adapted witness is a good instrument for movement, outline and direction, and a poor one for a face.

END OF THE DAY 18W

An account is not collected; it is produced, by a method, in a room, with people in it.

The photograph arrives

PLAN CARD 130W

Monday of the second week. Solomon has three documents laid side by side, with their dates showing. The account from 18 November. The evening paper from the next day. And the statement Cardoza signed on 26 December. Tomas Weill, the psychometrician, has already said what interests him. It is not what any of them says. It is the order they were written in. Ines Baptiste wants to know if the review is calling the witness a liar. Ren Okada has spent the morning explaining why that is the wrong question. Today you work out what the arrival date of a detail tells you about where it came from. A frightening night can do a lot to a memory. It cannot plant a detail that first shows up five weeks later.

BRIEFING 14W

The Hallam Register ran Ward's photograph on 19 November, the day after the interview.

WHAT YOU DECIDE 11W

Find out what entered the account after the night it describes.

6.1 The scar

MEMORY & INTERVIEW · INTERVIEW SUITE A · A ROOM · DIAGNOSIS

SCENE 37W

Ren Okada has four dated items on the screen when Unit Director Ines Baptiste asks whether Cardoza is lying about the scar. Okada says that question arrives too late. He puts the dates across the corridor instead.

GUIDE 58W

Five items, and four of them carry dates. Ask of each candidate whether the dates support it. The long first interview contains no scar. Ward's photograph appears the next day. The scar arrives later. Memory keeps content better than it keeps where the content came from, so an honest and confident memory can still have the wrong source.

ASK 12W

Which explanation fits every dated item rather than the most damning one?

BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

TAKES AS READ

a remembered detail carries no label saying where it came from

VERDICT 78W

Memory often preserves content better than source. A detail seen in a newspaper can later feel as though it came from the event itself, especially after repeated retrieval. The dates matter here: Cardoza's first full interview contains no scar; Ward's photograph appears the next day; the scar appears later. That chronology makes the photograph a plausible source of the new detail. It does not prove the photograph caused it, and later confidence cannot repair the missing source information.

TAKEAWAY 28W

Post-event information can later be remembered without a reliable source tag, so a new detail must be dated before it is treated as part of the original event.

6.2 Close the forty-two days

TESTING & RECORDS · DATA ROOM · A PERSON · BALANCE

SCENE 37W

Tomas Weill lays four dates along the identification corridor: 14 November, 18 November, 19 November and 26 December. The spans almost close. Nell Achterberg has independent records for each date and one missing interval to account for.

GUIDE 59W

Reading a record costs nothing, so read them all. Then count only the intervals that belong in the same ledger — the forty-two days between the first date and the last. One interval is missing, and finding where it goes is the point: the answer changes which sources the witness could have been exposed to before her certainty statement.

ASK 18W

Where do all forty-two days go, and which missing interval changes what sources were available to the witness?

BEHIND THE BUTTON 126W

Why the days have to close. Four dates and the spans between them should account for the whole period. If they do not, some days are unaccounted for, and unaccounted days are days when something could have happened that nobody has recorded.

What an interval means here. Not idle time. Time during which a witness may have seen a photograph in a newspaper, spoken to another witness, or been told that somebody had been charged. Each of those changes what a later identification is evidence of.

Why independent records matter. Achterberg's records were made for other purposes — custody, duty rosters, correspondence — so they were not written to support anybody's account. That is what makes them able to close a gap the case file cannot.

TAKES AS READ

a document's date can be checked against the record of when it was received · encoding: memory is built at the moment, from what was attended to — taken as read

VERDICT 79W

Chronology is evidence only when its intervals are independently anchored. The robbery is 14 November. The full interview is four days later, the photograph one day after that, and the array fifteen days later. The later certainty statement follows after another twenty-two days. Those intervals close to forty-two days. Leaving out the one-day newspaper interval creates a gap exactly where new face information entered. Dating does not prove causation, but it defines which sources were available before each account.

TAKEAWAY 21W

Chronology is only as complete as the intervals it preserves, especially when a missing interval contains a new source of information.

6.3 What the hormones cannot explain

STRESS, SLEEP & THE BODY · PHYSIOLOGY BAY · A ROOM · CHOICE

SCENE 33W

Raghunathan is asked whether the stress of the robbery could have delayed the scar into the account. She draws the time course of the alarm response on the board: seconds, minutes, and hours.

GUIDE 59W

Four options, and the question is a window in time. Ask of each when the mechanism it names is actually acting. The alarm response works during the event and the hours after it. Five weeks is long outside that. Stress still explains uneven encoding across the scene, which is a different claim about a different part of the file.

ASK 17W

Why can the stress response not explain a detail that first appears five weeks after the event?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

the stress response has a time course that can be measured

VERDICT 79W

Stress biology acts during the event and the hours that follow. The amygdala responds rapidly, and cortisol changes over minutes. Those systems can affect attention, encoding and consolidation. They cannot insert a new facial feature five weeks later. The relevant biological window has long closed by then. Stress still matters elsewhere in this case because it can explain uneven encoding across the scene. It does not explain a detail that first appears after later information has entered the record.

TAKEAWAY 21W

Every mechanism has a window it operates in, and a cause has to reach the moment it is being blamed for.

6.4 How much light the corner had — Review

PERCEPTION & THE SCENE · RECONSTRUCTION BAY · A ROOM · CALLBACK · CHOICE

SCENE 31W

Frey has moved the bay's lamp to the shop window and asks the same question about a different source. The window is twelve metres from the doorway on the other side.

GUIDE 60W

Four answers, and three of them move the light because the observer moved. Ask what illuminance is a property of. It belongs to the surface and the source, and neither of them walked anywhere. What her ten metres changes is the angle the face subtends, which is a separate calculation. Mixing the two is the commonest error in this material.

ASK 18W

A witness steps ten metres further back from the doorway. What happens to the light on the doorway?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

illuminance depends on the distance from the source to the surface being lit

VERDICT 74W

Illuminance is a property of the surface and the source. Moving an observer changes neither. What her extra ten metres changes is the angle the face subtends, and that is a separate calculation. Putting the observer distance into the light arithmetic is the commonest error in this material. The two belong apart because they cost the identification different things. One takes detail out of the scene. The other takes it out of the eye.

TAKEAWAY 17W

Illuminance belongs to the surface and the source; where anybody stands to look is a separate calculation.

END OF THE DAY 22W

A detail that is absent before a photograph and present after it has a source, and the source is not the night.

Two witnesses, one source

PLAN CARD 104W

Tuesday. The strongest looking page in the file is the one that lines the two witnesses up. Deniz Aktaş was behind the counter that night. His December statement agrees with Cardoza's on the jacket, the build, the way the man ran and the mark on his chin. Marcus Ilori has been saying for a week that agreement is the easiest thing in the world to make. Today you open each account and find out what it was built from. Then you work out what Aktaş could actually see through his own window. Elias Ward was convicted on two accounts. The two accounts had met.

BRIEFING 9W

The shop assistant's statement matches Cardoza's in four particulars.

WHAT YOU DECIDE 11W

Find out whether the agreement between the accounts is worth anything.

7.1 Agreement is not independence

INFLUENCE & THE ROOM · DELIBERATION ROOM · A ROOM · TRACE

SCENE 38W

Five items in the file point the same way. Grace Ntuli has them on the board with their derivations closed. Ines Baptiste asks why the unit is spending a morning on the part of the case nobody disputes.

GUIDE 54W

Open each of the five items to see what it was derived from. Keep the ones that stand on their own, untick the ones whose chain runs through a shared source, and then name that source. Four items pointing the same way is not four pieces of evidence if all four came from one.

ASK 24W

Four of these five point at Ward. Open them, keep whatever still stands on its own, and name the source they have in common.

BEHIND THE BUTTON 121W

Why the undisputed part is worth a morning. A case that everybody agrees about is a case nobody has audited. The four items agree because they are derived from one another, and the agreement is what made them look unassailable.

How dependence hides. A statement quoted in a summary, then cited by a report, then relied on in a later interview, appears three times in the file and is one observation. Closing the derivations is the only way to see it.

What survives, and why it matters. Anything that came from a separate observation still stands after the shared source is removed. The case does not disappear; it becomes accurately sized, which is the difference between a review and a demolition.

TAKES AS READ

two pieces of evidence built on the same source can only ever agree

VERDICT 73W

Agreement becomes stronger evidence when the sources could have failed differently. Three items here pass through the same newspaper photograph, so their later agreement is partly dependent on a common source. That does not make the details worthless; it means they cannot be multiplied as independent corroboration. The till roll and depot camera are different because neither depends on the photograph or on a witness identifying Ward, so they retain separate evidentiary weight.

TAKEAWAY 14W

Shared sources reduce independence; agreement can remain informative without counting as multiple independent confirmations.

7.2 Twenty minutes on the kerb

MEMORY & INTERVIEW · INTERVIEW SUITE A · A ROOM · TRACE

SCENE 35W

The first officers arrived at 21:58. Between the robbery and their arrival, Cardoza and Aktaş stood together outside the shop. Okada asks what the file can now say about the four particulars they agree on.

GUIDE 55W

Six statements in the file, and five sources they could rest on. Name the source the two accounts share, then keep only the statements that do not come through it. Two witnesses agreeing is two pieces of evidence only if they could have disagreed — and for twenty minutes on the kerb they could not.

ASK 15W

Which statements in the file are still independent of the twenty minutes on the kerb?

BEHIND THE BUTTON 144W

Why agreement is not independence. Once two people have discussed what happened, each account contains the other. The agreement is then a fact about the conversation rather than about the robbery, and it feels like corroboration to everybody involved, including the witnesses.

Why it cannot be unpicked afterwards. What one witness heard from the other is encoded the same way as what she saw herself, with no tag saying which is which. Asking her to separate them asks for something the memory system does not store, which is why the only remedy is separating witnesses before they talk.

What survives. Anything each of them recorded before the twenty minutes — the triple-nine call, a text message with a timestamp — and anything only one of them could have known. Those are genuinely two sources, and they are what the file can still add together.

TAKES AS READ

people compare accounts when something has just happened to both of them

VERDICT 81W

Two accounts are strongest as corroboration when they could have come out differently. Cardoza and Aktaş spoke together for roughly twenty minutes before their later formal statements, so details they discussed acquired a shared pathway. That is co-witness contamination: later agreement on those details is not clean evidence of two independent memories. It does not prove the details are false. The pre-conversation emergency call and timestamped text remain especially valuable because each was recorded before the witnesses could influence one another.

TAKEAWAY 21W

Once witnesses exchange details, later agreement on those details is no longer independent corroboration; records made before they spoke still are.

7.3 Looking out of a lit room

PERCEPTION & THE SCENE · RECONSTRUCTION BAY · A ROOM · VERIFY

SCENE 35W

Bekele has the bay set up from the shop side. The counter, the window, the interior lights at the level the shop record gives. Beyond the glass, the street at the illuminance the corner had.

GUIDE 65W

Say what the counter position can resolve on the street after two minutes of looking out, then run the bay and read the result. The prediction is the part that matters: the eye behind the counter is set for the shop, not for the pavement, and how far along the adaptation curve two minutes gets is a number worth committing to before the answer appears.

ASK 20W

Predict what the counter position can resolve out on the street after two minutes, then run the bay and measure.

BEHIND THE BUTTON 177W

What dark adaptation is. Cone vision adjusts within about a minute; rod vision takes twenty to thirty and gives no colour and poor acuity. An eye that has been in a lit shop is at the very start of that curve when it turns to a dark street, so the first minutes are the worst of both — cones short of light, rods not yet in.

Why the glass makes it worse. A lit interior reflects in the window, adding a veil of light that never came from outside. That veil raises the level the eye is adapting to and reduces the contrast of everything beyond it, which is why cupping hands to the glass is what people do without being told.

Why two minutes is the honest number. The witness statement puts the man at the counter for a couple of minutes before the shout, so the reconstruction has to model that eye, not a fully dark-adapted one. The difference is not subtle: a few minutes on the curve is a factor of several in threshold.

TAKES AS READ

the eye adjusts to the general level of light it is in

VERDICT 197W

Two visual systems with different clocks are what makes this answerable. Cones adapt over the first several minutes, while rod sensitivity continues improving for roughly twenty to thirty minutes. Dim scenes become increasingly rod-dominated, with poorer colour discrimination and coarser acuity. An eye that has spent the evening in a lit shop is at the beginning of that curve, so at two minutes it is short of both. Measured in the bay at the corner's illuminance, the counter position resolves detail of about 0.9 m — a person, a direction, a gait, and no face. After twenty minutes rod sensitivity is much greater, though spatial detail remains coarse; the reconstruction's fully adapted result is therefore a different viewing state, not a correction that can be applied automatically to the witness. The window adds the

rest: the lit interior reflects in the glass and lays a veil over everything beyond it, raising the level the eye is adapting to and cutting the contrast of the street. That is why the reconstruction has to model the eye the witness actually had, and why predicting first is the discipline — a threshold looked up afterwards fits whatever the bay produced.

TAKEAWAY 21W

An eye set for a lit room is the wrong eye for a dark street, and the glass makes it worse.

7.4 A word where a reading should be — Review

TESTING & RECORDS · DATA ROOM · A ROOM · CALLBACK · CHOICE

SCENE 34W

Achterberg hands over a page from another case. It records that a corridor was "well lit", that a door was "about two metres" from a bed, and that a lamp was "the usual type".

GUIDE 60W

Three entries from another file, and four options. Ask of each entry whether it says what would count as disagreement. A quantity and a unit do, even a rough one. Well lit and the usual type do not, however sincere the writer was. So the test is not whether an entry sounds precise but whether somebody else could contradict it.

ASK 10W

Which of these three entries is closest to a measurement?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a measurement states its units and can be repeated

VERDICT 93W

A measurement names a quantity and a unit, which is what lets somebody else go back and get the same number or a different one. "About two metres" is a weak measurement — no instrument, no tolerance — and it is still checkable in a way that "well lit" and "the usual type" are not, because it says what would count as disagreement. That is the test to apply to any file: not whether the entry sounds precise, but whether it says what would have to be true for it to be wrong.

TAKEAWAY 17W

A quantity somebody could repeat is a measurement; everything else is a description, however numerical it looks.

END OF THE DAY 16W

Two accounts that share a source are one account, and the file counts them as two.

Did the array point at Ward?

BEFORE THE DAY — HUNT

Six pieces of the 2019 file, and the box holds four

The array form, the custody log, the two contemporaneous statements, the lux survey and the officer's notebook. Four are in the box. Until the other two are found the review cannot say whether the procedure was followed or whether the record of it is simply missing, and those are different findings.

PLAN CARD 106W

Wednesday. The mock witness run finished overnight. Forty volunteers read the description Cardoza gave on the night of the robbery. Then they saw the 2019 array of photographs and pointed at whoever fitted. Not one of them has ever seen the robbery. Marcus Ilori has the tally. He will not say the number out loud yet. First somebody has to work out what a fair number would be. Today you turn that tally into a size. Then you decide what would make the procedure a sound one. Last, you look at what a photograph asks an eye to do. Ward was picked out of six faces.

BRIEFING 10W

Forty volunteers have read Cardoza's description and seen nothing else.

WHAT YOU DECIDE 18W

Test whether the six-photo array offered real alternatives and identify reforms supported by evidence rather than by habit.

8.1 Six photographs, and how many alternatives

IDENTIFICATION PROCEDURE · IDENTIFICATION SUITE · A ROOM · BALLPARK

SCENE 35W

Marcus Ilori spreads forty mock-witness tally marks across the identification bench. Twenty-six sit under one photograph. The volunteers had Cardoza's description and nothing else. He asks how many real alternatives the six-photo array actually offered.

GUIDE 55W

Five numbers, and three belong elsewhere: the photographs, the fillers, and the volunteers in the other bay. Ask of each whether this measure needs it. Protection comes from real alternatives rather than printed ones. Mock witnesses know only the description, so where they land is a measurement of the array rather than of the case.

ASK 8W

Estimate the functional size of the 2019 array.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a lineup member who nobody would pick from the description is not an alternative · how a lineup is built: fillers, and the ones nobody would pick — taken as read

VERDICT 79W

Lineup protection comes from real alternatives, not from the printed number of photographs. If all six fit the description equally, guessing should spread across them. Mock witnesses test that because they know only the description. Here twenty-six of forty choose Ward's photograph. Functional size is therefore 40 divided by 26, about 1.54. The nominal array had six members. Its effective protection was closer to one and a half alternatives, making an innocent-suspect pick far easier than the form suggests.

TAKEAWAY 22W

Nominal size is how many faces are on the table; functional size is how many of them a stranger could land on.

8.2 Moving the middle is not narrowing the spread

TESTING & RECORDS · DATA ROOM · A ROOM · CLOUD

SCENE 34W

Weill has the mock-witness result as a distribution against the band a fair six-member array would sit in. Four changes are available to the unit's recommendation and each of them costs days of drafting.

GUIDE 61W

The cloud is what this procedure produces across administrations; the band is where a fair six-member array would land. Four changes are available and each costs days of drafting. They act differently — some move the cloud, some tighten it — so watch the middle and the spread separately. Apply what you would recommend, then say when the procedure is defensible.

ASK 25W

Bring the recommended procedure inside the fair band, and say when it is defensible. Place the bars to report the mean and its one-sigma uncertainty.

BEHIND THE BUTTON 163W

What the two shapes are. The band is a standard: run a fair array many times and this is the share of mock witnesses who land on the suspect. The cloud is this unit's own procedure measured the same way. A procedure whose cloud sits outside the band is producing identifications a fair array would not.

Moving and tightening are different fixes. Matching fillers to the description or adding a may-not-be-present caution changes how strongly the array points toward the suspect. Blind administration and a standard script remove avoidable administrator and site variation. This authored cloud is a teaching model of those different failure modes; it does not assign a universal effect to simultaneous or sequential presentation.

Why 'defensible' is the word. This becomes a recommendation other units follow, and it will be read out in court. That makes the claim a claim about every administration rather than about the average one, which is why the criterion is written on the whole cloud.

TAKES AS READ

with equally plausible members and conditional on a mock witness making a choice, a fair six-member array sends about one choice in six to the suspect · a recommendation can propose changes to a procedure as well as criticise it

VERDICT 65W

A fair procedure needs both a defensible average and a controlled way of producing it. Better fillers and a may-not-be-present instruction address how strongly the array points toward the suspect. Blind administration removes a pathway for expectancy, and a standard script reduces site-to-site variation. This authored simulation shows the four protections working together. It does not establish that any one presentation format is universally superior.

TAKEAWAY 20W

Improve the parts of a lineup with well-supported failure modes—fillers, instructions, blinding and documentation—without claiming one presentation format always wins.

8.3 A face on paper

PERCEPTION & THE SCENE · RECONSTRUCTION BAY · A PERSON · CHOICE

SCENE 36W

Frey has the array photographs and a live volunteer at the same distance under the same lamp. She asks the room which of the two tasks is closer to what Cardoza was doing on 14 November.

GUIDE 55W

Four things a photograph array removes. Ask of each whether real recognition depends on it. A remembered face was never a still image. Head turns, expression and gait all add information across time, and a print freezes one angle under one light. That also makes the photographs uneven in a way no filler rule controls.

ASK 11W

What does an array of photographs remove from the recognition task?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a face is recognised from more than a single frozen view of it

VERDICT 77W

Real face recognition draws on changing views. Head turns, expression, gait and feature motion all add information across time. A lineup photograph freezes one angle under one lighting condition. The witness must compare that single view with a memory that was never a still image. That makes the task harder and can make photographs uneven. One person's chosen image may resemble the remembered view more by chance than another person's, even when both are equally plausible fillers.

TAKEAWAY 21W

A stand-in for a task is easier or harder than the task itself, and the difference is whatever it leaves out.

8.4 A filler, or a giveaway

IDENTIFICATION PROCEDURE · IDENTIFICATION SUITE · A ROOM · BELT

SCENE 36W

The array has to be built from the photograph library before the witness arrives. A filler works if somebody who never saw the offender could plausibly pick it. Anything else narrows the array without anybody noticing.

GUIDE 51W

Two bins. A photograph works as a filler if it fits the witness's own description as well as the suspect does. It fails if it stands out — wrong description, different background, a mugshot board, or simply the only one of something. Sort on whether a stranger could plausibly pick it.

ASK 14W

Send each photograph to the bin that says whether it works as a filler.

BEHIND THE BUTTON 80W

Why a bad filler is worse than a missing one. An array of six with three implausible fillers behaves like an array of three, so the chance of a false identification rises and nothing in the paperwork records that it did.

What makes a photograph stand out. Not being like the suspect — being unlike the others. A different background, a visible custody board, the only person with a beard when the description said clean-shaven, the only photograph in colour.

TAKES AS READ

a lineup contains one suspect and people known not to be the offender

VERDICT 154W

A lineup's protection comes from its effective size, not its nominal one. Six photographs where three could never be chosen is an array of three, and the chance that a guessing witness lands on the innocent suspect goes from one in six to one in three — with the form still saying six. What makes a filler fail is standing out from the others rather than differing from the suspect: a different background, a custody board visible at the bottom, the only one in colour, the only one photographed from slightly above. And the description is the test that matters. If the witness said clean-shaven and mid-thirties, then every filler has to be clean-shaven and mid-thirties, because a witness scanning for the description eliminates anybody outside it before making a choice at all. That elimination is invisible afterwards, which is why the array is scored when it is built rather than argued about later.

TAKEAWAY 15W

A filler that nobody would pick is not in the array, whatever the paperwork says.

END OF THE DAY 15W

A lineup's real size is how many of its members a stranger could plausibly pick.

Thirty-four metres

PLAN CARD 113W

Thursday of the second week. Two numbers are up on the same wall. Dawit Bekele has surveyed Ferrier Street. His survey puts the doorway thirty-four metres from the spot where Cardoza says she stood. The case summary says twenty-two, and has said so since 2020. Nothing on that street has moved since the robbery. So one of these numbers is a measurement. The other one is not. Today you find out which is which. Then you work out what the eye can pick out at the distance that survives. Last, you decide what the review is allowed to say about the moment she picked Ward out. Everything the file argues rests on this.

BRIEFING 21W

The number the case has trusted for seven years is wrong. The file says twenty-two metres; the independent geometry says thirty-four.

WHAT YOU DECIDE 12W

Settle the distance, and work out what it leaves of the identification.

9.1 Where the twenty-two came from

TESTING & RECORDS · DATA ROOM · A ROOM · DIAGNOSIS

SCENE 36W

Bekele hooks the tape at Cardoza's repeated position and walks past the file's twenty-two-metre floor mark. The tape keeps going to 34.2 m. Tomas Weill puts the unscaled sketch and the shop lease plan beside it.

GUIDE 58W

Five items, and two of them are independent measurements that agree with each other. Ask of each candidate how many of the five it fits. The witness places herself in the same spot in three accounts. The lease plan matches the survey. The one figure with nothing repeatable behind it traces to a sketch marked not to scale.

ASK 8W

Which explanation is consistent with all five items?

BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

TAKES AS READ

a measurement can be repeated and an estimate cannot

VERDICT 77W

The 22-metre figure is the only distance here without a repeatable measurement behind it. Bekele's survey gives 34.2 m, and the lease plan independently matches that geometry within 0.2 m. Cardoza also places herself at the same location across three accounts. The summary traces back to a sketch marked not to scale. Repetition then gave the estimate the appearance of a measured fact. Nothing has to be falsified for a number to acquire false authority through copying.

TAKEAWAY 18W

A number that entered a file without a measurement behind it never becomes a measurement by being repeated.

9.2 A centimetre of face

PERCEPTION & THE SCENE · RECONSTRUCTION BAY · A PERSON · BALLPARK

SCENE 37W

Frey moves the mark to the surveyed distance and reads out the eye's resolution limit in good light. She wants to know what size of detail that limit corresponds to at the far end of the bay.

GUIDE 57W

Five numbers, and three of them belong to other stops: the earlier distance, the eye gap, and a light level. Ask of each whether this limit needs it. An angular limit becomes a size only once a distance is supplied. And note the order of the argument: distance shrinks the detail before lighting is considered at all.

ASK 12W

Estimate the smallest detail an eye could resolve at the surveyed distance.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

an eye in good light resolves details about one arc minute across · one arc minute is about 0.0003 radians

VERDICT 79W

An angular resolution limit becomes a physical size once distance is supplied. At 34 metres, 0.0003 radians corresponds to about 0.0102 m, roughly one centimetre under good high-contrast conditions. Fine facial differences often live at that scale or below, and low illumination can make practical discrimination worse. The corrected distance therefore reduces the available facial detail before the historical lighting question has even been settled. The maintenance record still has to answer what light actually belonged in the reconstruction.

TAKEAWAY 17W

The resolution limit is an angle, so the size of the smallest visible detail grows with distance.

9.3 What this does and does not settle

IDENTIFICATION PROCEDURE · IDENTIFICATION SUITE · A ROOM · ATTEST

SCENE 38W

Baptiste wants the finding in the report by the end of the week. Ilori asks her to write the sentence first, because the sentence is where reviews are lost, and the room spends an hour on one line.

GUIDE 66W

Six statements the draft could make, and three verifications. Each rests on something in the file rather than on something anybody has seen, so check the ones that would hurt most if the file were wrong. A statement can only be held once it has been verified — and two of these cannot be supported by any measurement of the corner, however carefully it was taken.

ASK 15W

Which statements in the draft cannot be held, given what the review's evidence actually supports?

BEHIND THE BUTTON 185W

What a limit on perception establishes. The measured distance, light and reconstruction can show that reliable facial identification was substantially harder than the file implies. That is a claim about evidentiary weight and the conditions. It does not establish who was on Ferrier Street, and no measurement of light or distance can, because absence of sufficient information is not information about the answer.

Why the distinction decides the sentence. "The measured conditions make this identification much less reliable than the file treated it" is what the review has evidence for. "Ward was not there" is a different claim, resting on evidence nobody in this building has, and a report that slides from the first to the second loses everything the first one earned.

What retrieval adds. A recognition is reconstructed at the moment of asking rather than replayed, so confidence can coexist with poor viewing conditions and later influence. That is compatible with the mechanism; it does not by itself prove how this identification arose. Confidence is not a check on the conditions, and a witness's certainty is not evidence that the conditions were adequate.

TAKES AS READ

a limit on what could be seen is not evidence about who was there · storage and forgetting: the curve is steep at the start — taken as read

VERDICT 91W

The review now has strong evidence for two kinds of claim: geometry and procedure. Bekele's survey gives 34.2 metres and the independent lease plan agrees; the 2019 array form lacks a may-not-be-present caution. The historical lighting is not yet settled, because a correct lamp calculation is not evidence that the lamp was on. And none of those findings identifies the person at the doorway. The review can say that

the case file overstated the strength of the identification without saying that Ward was absent or that Cardoza was dishonest or mistaken.

TAKEAWAY 16W

A measurement can show that an identification deserves less weight without showing who was actually there.

9.4 The order an account is taken in — Review

MEMORY & INTERVIEW · INTERVIEW SUITE A · A ROOM · CALLBACK · CHOICE

SCENE 35W

Okada counts the times Cardoza was asked about the chin between November and the trial. The answer is nine, by five different people, four of whom already knew what the answer was expected to be.

GUIDE 60W

Four accounts of what nine askings do. Ask of each whether it is about the witness's patience or about what the repetition tells her. A question asked again says somebody was not satisfied. Adults shift under that, and children shift far more. So every version after the first was produced under that pressure, and nobody here had counted the askings.

ASK 12W

Why does asking about the same detail nine times damage the account?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a question can be asked more than once about the same detail

VERDICT 97W

A question asked again carries information: somebody was not satisfied. Adults shift under that pressure. A child witness shifts far more, and suggestibility rises with each repeated question, which is why guidance on interviewing children is largely a set of rules about not asking twice. The account does not decay so much as drift toward whatever the repeated asking implies, and every version after the first is a version produced under that pressure. It is also why the count of how often a question was put belongs in a review, and why nobody had counted it here.

TAKEAWAY 18W

Repeating a question implies the first answer was wrong, and the account moves toward whatever the asking suggests.

END OF THE DAY 18W

The most important number in a case can be wrong for seven years because nobody ever measured it.

What did the administrator change?

PLAN CARD 106W

Friday. Karen Loomis, the retired detective, has come in for the second time. She ran the photo array in 2019. She knew which of the six photographs was Ward. She remembers saying something to Cardoza afterwards. She does not remember what it was. Four things about that afternoon could have mattered. The file cannot pull them apart. Today you take them apart one at a time, in the unit's own rig. Then you work out how a belief travels from an officer to a witness with neither of them noticing. Last, you read what praise does to an account. Ilori trained the officers who replaced Loomis.

BRIEFING 14W

Loomis knew which photograph was the suspect, because in 2019 the administrator always did.

WHAT YOU DECIDE 9W

Establish what the administrator's knowledge did, by removing it.

10.1 Change one thing

IDENTIFICATION PROCEDURE · IDENTIFICATION SUITE · A ROOM · CONTROL

SCENE 38W

The unit's rig runs twenty volunteer administrations at a time. Four features of the 2019 procedure can be switched independently. The readout is reported certainty after a choice, and the room has to isolate which feature moves it.

GUIDE 58W

The number you are watching is the certainty volunteers report after they have chosen. Four features of the 2019 procedure can be switched independently, and twenty administrations run in a block. A response counts only if it is larger than the block-to-block scatter. Switch one feature, run a block, switch it back, and name the feature certainty follows.

ASK 16W

Four features, one effect on certainty. Establish which of them is producing it and prove it.

BEHIND THE BUTTON 166W

Why certainty is worth measuring at all. It is what a courtroom hears: a witness who says they are sure is believed more than one who hesitates. If a feature of the procedure can move certainty without changing whether the choice was right, then the procedure is manufacturing confidence, and the court has no way to tell that from the inside.

What the four features are. Whether the administrator knows which photograph is the suspect, whether the six are shown together or one at a time, whether the witness is told the man may be absent, and where the block is run. Each is a real difference between police procedures in use, and each is defensible until it is tested.

Why one at a time, and why put it back. Four features switched together give one number and no attribution. Switching back guards against the other explanation: volunteers who arrived later in the day, or a room that got noisier, drifting the number while you worked.

TAKES AS READ

a feature can only be shown to matter by changing it and changing it back

VERDICT 81W

Several features changed together in the original procedure, so the case file alone cannot identify a cause. A controlled volunteer test changes one variable while holding the others fixed, and a reversal checks whether the response follows that variable in both directions. In this authored rig, administrator knowledge moves the outcome. The case-level conclusion is deliberately narrower: because Loomis knew the suspect, a known pathway for expectancy was open. The experiment cannot measure how strongly that pathway affected Cardoza in 2019.

TAKEAWAY 23W

A controlled experiment can establish a mechanism; a historical file still has to show whether and how strongly that mechanism reached the case.

10.2 Nobody said anything

INFLUENCE & THE ROOM · DELIBERATION ROOM · A ROOM · CONTROL

SCENE 35W

Loomis is clear that she did not point, prompt or name anybody. Ntuli believes her and plays a recording of a volunteer administration where the same thing is true and the effect is visible anyway.

GUIDE 60W

Four things differ between the administrations where the witness picked the suspect and the ones where she did not. Change one and run the volunteer session again; the readout is how often the suspect is chosen. Changing several at once tells you nothing about any of them, and putting your suspect back is what separates a cause from a coincidence.

ASK 14W

Which difference between the sessions is actually moving how often the suspect is picked?

BEHIND THE BUTTON 151W

What expectancy is made of. Not prompting or pointing, which nobody here did. It is carried in latency — the half-second before "take your time" — in which photographs get a second look, in the pitch of an acknowledgement, and in what happens after a choice rather than before it.

Why an honest administrator transmits it. The cues are not chosen and mostly not noticed, by either party. An administrator who knows which photograph is the suspect produces them; one who does not cannot, whatever their intentions. That asymmetry is the whole argument for double-blind administration.

Why the remedy is structural. You cannot train somebody out of a behaviour they are not performing deliberately, and you cannot audit it from a transcript, because the transcript records words. Removing the knowledge removes the mechanism, which is why the rule is about who runs the array rather than about how they run it.

TAKES AS READ

people take up small cues from somebody running a procedure they are taking part in

VERDICT 82W

The volunteer sessions isolate administrator knowledge while holding the photographs, witness pool, recording and time of day fixed. In this authored rig the suspect-pick rate follows that knowledge when it is removed and restored, supporting a causal effect in the experiment. That is why blind administration is a robust procedural protection. The 2019 record cannot tell the unit how much expectancy, if any, changed Cardoza's particular choice, so the report should describe an avoidable pathway rather than invent a case-specific effect size.

TAKEAWAY 20W

Blind administration closes an expectancy pathway; a historical review should not turn a laboratory effect into an unmeasured case-specific number.

10.3 What "that's our man" moves

MEMORY & INTERVIEW · INTERVIEW SUITE A · A PERSON · CHOICE

SCENE 35W

Ren Okada puts the unit's replication on the confidence screen. Half the volunteers are told their choice matched the suspect. Both groups then report how good their original view was and how long they looked.

GUIDE 54W

Four things feedback might move. Ask of each whether it is a claim about the night or about the present. Confirming feedback changes what witnesses report about the original view: how clear it was, how long they looked, where they stood. Those sound like evidence about encoding and are partly reconstructed after the choice.

ASK 12W

Besides certainty, what does confirming feedback change in a witness's later account?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a witness reports the conditions of an event from memory, like everything else

VERDICT 76W

Post-identification feedback changes more than stated certainty. After being told their choice matches the suspect, witnesses often report a clearer view and longer attention at the event. They can also remember a better position for seeing. Those claims concern the original night but are reconstructed after feedback. That matters in court because 'I had a clear look' sounds like direct evidence about encoding. It may instead partly reflect information learned after the identification was already made.

TAKEAWAY 22W

What somebody says about the circumstances of an event is itself remembered, so it can move long after the event has finished.

END OF THE DAY 20W

Four things wrong at once is not four findings; it is one until somebody changes them one at a time.

The streetlamp was out

PLAN CARD 103W

Monday of the third week. It is the quietest day the review has had. Nothing has gone wrong. Three things have been confirmed. The custody record settles the date of the certainty statement. The mock witness rates are in from both procedures. And the power company has answered Yara Solomon's letter about column 4471. Tomas Weill wants the two procedures compared properly. Not by which one produced more picks. Today you separate seeing from choosing. Then you settle when Cardoza's certainty was written down. Last, you read what the power company says about that lamp. One of those three findings changes the case.

BRIEFING 24W

The distance was worse than the file said. Today the second load-bearing claim breaks: the streetlamp was out on the night of the robbery.

WHAT YOU DECIDE 13W

Date the certainty, and separate telling faces apart from being willing to choose.

11.1 Seeing, and being willing to say so

TESTING & RECORDS · DATA ROOM · A ROOM · BALLPARK

SCENE 28W

Weill has two procedure conditions on the board. The 2019-style condition produced more identifications than a reformed blind condition, and Baptiste asks whether “more identifications” means “better identification.”

GUIDE 52W

Five numbers, and two of them are the raw rates rather than their standard scores. Ask of each what the measure is built from. A procedure can produce more identifications simply by making people readier to choose, which lifts hits and false alarms together. So more choices is not automatically better performance.

ASK 7W

Estimate the sensitivity of the 2019-style procedure.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a procedure can be run with the offender present and with the offender absent · a proportion can be converted to a standard score from a table

VERDICT 76W

A procedure can produce many identifications simply by making witnesses more willing to choose. That raises hits and false alarms together. Sensitivity separates this response tendency from actual discrimination. Converting both rates to standard scores gives a common scale. Here 0.52 minus negative 0.52 gives $d\text{-prime} = 1.04$. The raw rate difference of 0.40 is not comparable in the same way. More choices are not automatically better performance when the extra choices include more false alarms.

TAKEAWAY 14W

A procedure that produces more identifications may only be producing more of both kinds.

11.2 Buy the confidence that still means something

IDENTIFICATION PROCEDURE · IDENTIFICATION SUITE · A PERSON · VALUE

SCENE 38W

Case analyst Solomon brings four sealed record pulls to the confidence desk. Two can be authenticated before the day's board packet closes. Somewhere in them are Cardoza's words on 4 December and her certainty statement on 26 December.

GUIDE 59W

Four sealed record pulls, two authentications before the packet closes. Open each and read what it would establish. Cardoza's words on 4 December and her certainty statement on 26 December are somewhere in them, and the gap between those dates is the point. Buy the records that can show what her confidence was before it had time to grow.

ASK 10W

Which records are worth authenticating before the board packet closes?

BEHIND THE BUTTON 127W

Why the date matters more than the content. Confidence rises after an identification, especially when a witness learns that others agree or that a suspect was charged. A certainty statement taken three weeks later is a measurement of a different thing from one taken at the time.

What authenticating buys. A record's value depends on when it was made and by whom, and a sealed pull is a claim about both until somebody checks. An unauthenticated record put in front of a board carries the weight of a document and the reliability of an assertion.

Why two rather than four. The packet closes today, and the choice between them is the judgement being tested: which two records, if genuine, most change what the board can responsibly conclude.

TAKES AS READ

a witness's certainty can change between the identification and a later statement

VERDICT 71W

Confidence has its strongest relation to accuracy when it is recorded immediately after an uncontaminated identification. Later certainty includes everything learned after the choice. That can include feedback, charging decisions and repeated commitment to the identification. The 4 December form preserves Cardoza's own words: 'maybe, number four.' The 26 December statement says 'absolutely certain.' Both are real records. They answer different questions because three weeks of post-identification experience lie between them.

TAKEAWAY 17W

Confidence is evidence about accuracy only when its timing and the information available before it are preserved.

11.3 Column 4471

PERCEPTION & THE SCENE · RECONSTRUCTION BAY · A ROOM · CHOICE

SCENE 38W

Vision scientist Frey lays the utility fault ticket for column 4471 beside the bay switch. It runs from 2 to 21 November 2019. She kills the modeled lantern; the doorway meter settles near two lux from shop-window spill.

GUIDE 47W

Read the fault ticket against the earlier calculation. The arithmetic described the lantern if it was working; the ticket covers the robbery night. What remains is shop-window spill at roughly two lux: dim, mesopic vision in which colour, contrast and fine facial detail are poorer, not absent.

ASK 16W

What does the fault ticket do to the twelve lux the review calculated on day four?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

illuminance falls off with the square of the distance from a source

VERDICT 89W

The 12.5-lux calculation described column 4471 if its lantern was working. The utility ticket shows it was out from 2 through 21 November, including the robbery on 14 November. The arithmetic was correct and the premise was false. Bekele's reconstruction leaves about two lux of shop-window spill at the doorway. That is dim rather than dark and lies in a mesopic regime where both rods and cones can contribute. Colour discrimination, contrast and fine facial detail are poorer than in the working-lamp model, but they do not simply disappear.

TAKEAWAY 19W

With the column out, what light the corner had came from a shop window twelve metres the other way.

END OF THE DAY 17W

A quiet day in a review is three confirmations, and one of them usually changes the case.

The nineteenth hour

PLAN CARD 108W

Tuesday. The custody log is the last major document nobody has read closely. Ward was arrested at 06:40 on 3 December 2019. At 01:20 the next morning, in the nineteenth hour of custody and with no recorded sleep opportunity, he agreed with an account put to him. Eight hours later, after sleep, he took it back. The log can establish the timing and conditions; it cannot read his mental state. Today you ask what conclusions remain defensible under a pessimistic range of cognitive control, then examine the minimised version he was offered. Last, you compare a witness's later account with the one recorded before years of public exposure.

BRIEFING 13W

Nineteen hours in custody, no lawyer, and an agreement retracted the next morning.

WHAT YOU DECIDE 13W

Read the interview that produced the only words the file has from Ward.

12.1 What survives the bad end of nineteen hours

STRESS, SLEEP & THE BODY · PHYSIOLOGY BAY · A ROOM · STRESS

SCENE 39W

Priya Raghunathan and clinical psychologist Oyelaran put the custody log beside Ward's 01:20 agreement. Baptiste wants a sentence the board can use. The panel instead asks which treatment of the admission still works if cognitive control was badly degraded.

GUIDE 49W

Move the assumption about Ward's cognitive control to its bad end, and watch which treatments of the 01:20 admission survive. A sentence the board can use has to hold at the pessimistic end of nineteen hours, not merely at the likely one. Choose the treatment that still works there.

ASK 18W

How should the review use Ward's 01:20 agreement if his ability to weigh distant consequences was badly reduced?

BEHIND THE BUTTON 154W

What nineteen hours does. Sustained wakefulness degrades the ability to weigh distant consequences against immediate relief long before it affects speech or memory of ordinary facts. A person can be lucid, cooperative and articulate while being much worse at judging what agreeing now will cost later.

Why the range is the question. Nobody can measure what Ward's control was at 01:20. The evidence gives a range, and a review that picks the middle of it has made an assumption it cannot defend. Testing at the bad end is what turns an untestable claim into a decidable one.

What survives, and what does not. Treatments that rely on the admission being a considered decision fail at the pessimistic end. Treatments that use it as one item among others, weighted by the conditions it was made in, hold — which is why the answer is about how to use it rather than whether to believe it.

TAKES AS READ

the ability to weigh a future consequence against a present one degrades without sleep

VERDICT 91W

Prolonged custody and the absence of a recorded sleep opportunity can impair attention and decision control, but the file does not measure Ward's exact cognitive state. The robust question is therefore not whether exhaustion "caused" a false statement. It is what evidentiary treatment remains safe across plausible levels of impairment. Ward supplied no unique detail and retracted the agreement after sleep without receiving new case information. Requiring independent corroboration for any detail he supplied is defensible even at the pessimistic end without treating fatigue as proof that the statement was false.

TAKEAWAY 21W

When a psychological state was not measured, use it as an uncertainty range rather than as a diagnosis that decides truth.

12.2 The cheap version

INFLUENCE & THE ROOM · DELIBERATION ROOM · A PERSON · BALLPARK

SCENE 34W

Ntuli reads the account that was put to Ward at 01:20. It describes a man who went along with something, who did not plan it, and who was not the one holding the bag.

GUIDE 74W

Work the arithmetic the room offered him. Two numbers were on the table at 01:20 — what the minimised version was presented as costing, and what holding out was presented as costing — and the expected years of the second one is the sentence times the chance of conviction he was given. One tile is the sentence the law actually attaches to the minimised account, which is the number nobody in the room said.

ASK 15W

What did holding out appear to cost him, as the room presented it at 01:20?

BEHIND THE BUTTON 175W

What minimisation does. It lowers the apparent cost of admitting, by offering a version of events that sounds partial, reluctant and forgivable. Maximisation raises the apparent cost of denying, by presenting conviction as near-certain. Together they change the arithmetic a person is doing without changing anything about what they did.

Why the arithmetic is what matters and not the person's character. For an innocent person, admitting a lesser version at an apparent nine months against an apparent near-certain eight years is a rational trade at that moment, in that room, on those numbers. That is the mechanism, and it does not require anybody to be weak or confused.

Why the numbers were not real. The law does not price a minimised account the way the interview room did: going along with a robbery in which a weapon was carried attracts a sentence in the same band as planning it. So the trade offered was between a real eight years and an imaginary nine months, and only one side of it was ever going to arrive.

TAKES AS READ

people admit more readily to a version that costs them less

VERDICT 187W

Minimisation and maximisation work on arithmetic rather than on character. At 01:20 the room presented two options with numbers attached. Hold out, and conviction was put at 90 per cent certain with a sentence of eight years: $0.90 \times 8 = 7.2$ expected years. Take the offered version — went along with it, did not plan it, was not holding the bag — and the implication was a matter of months. On those numbers, admitting is the rational move for a guilty person and for an innocent one alike, which is the whole point: the mechanism does not need the person to be weak, confused or unintelligent. What makes it a false-confession risk is that one of the

two numbers was not real. The law does not price a minimised account the way an interview room does, and going along with a robbery where a weapon was carried attracts about six years — so the trade on the table was a genuine 7.2 against an imaginary 0.75, and the reason it looked like a good deal is that the second number was never going to be honoured.

TAKEAWAY 21W

Minimisation lowers what an admission appears to cost, and the law does not price it the way the interview room does.

12.3 What Aktaş remembers now

MEMORY & INTERVIEW · INTERVIEW SUITE A · A ROOM · VERIFY

SCENE 35W

Baptiste asks whether the unit should re-interview Cardoza before the report closes. Ren Okada puts Deniz Aktaş's 2023 interview beside his 2019 statement. Four new details appear, each already present somewhere in the public record.

GUIDE 68W

Say what fraction of the 2019 account a re-interview four years later should still reproduce, then run the comparison and read the result. The curve is steep at the start and shallow later, so most of what was going to be lost went in the first weeks. Commit to a number before the answer appears — the interesting finding is not how much was lost but what arrived.

ASK 16W

Predict how much of the 2019 account the 2023 interview still reproduces, then run the comparison.

BEHIND THE BUTTON 166W

What the forgetting curve does. Retention falls fast at first and then flattens, so the difference between four years and four months is much smaller than the difference between four months and four days. An account four years old is not four years' worth of decay; it is a few weeks' decay and then a long plateau.

Why the new details are the finding. Four particulars appear in 2023 that are not in the 2019 statement, and every one of them is somewhere in the public record — the paper, the appeal coverage, the campaign. Memory does not tag a detail with where it came from, so they are held with the same confidence as everything else.

What follows for the decision. A fresh interview cannot recover the 2019 account, because nothing subtracts what has arrived since. It can only produce a 2023 account, and the unit already has one. The 2019 statement is the evidence; a re-interview adds a document about the intervening four years.

TAKES AS READ

an account can be compared with an earlier account by the same person

VERDICT 187W

The forgetting curve is steep at the start and shallow afterwards, so most of what was going to be lost from the 2019 statement was gone within weeks and the four years since have cost relatively little. Compared item by item, about seven particulars in ten survive intact — which is roughly what the curve predicts and is not the finding. The finding is the other direction. Four details appear in 2023 that are not in the 2019 statement, and each one can be found in the newspaper report, the appeal coverage or the campaign material. Memory stores no tag saying where a detail came from, so Aktaş holds those four exactly as he holds the ones he saw, with the same confidence and no sense of having acquired them. That is why the answer to Baptiste's question is no. A re-interview cannot recreate the 2019 memory state,

because nothing subtracts the intervening four years. It produces a 2023 account whose differences may reflect forgetting, later information, retrieval and ordinary variability; tracing new details to public sources makes post-event influence a plausible explanation rather than a certainty.

TAKEAWAY 23W

A later account includes years of forgetting, retrieval and post-event exposure; it cannot simply be subtracted back to recover the 2019 memory state.

END OF THE DAY 20W

An admission is evidence about the interview as well as about the event, and the interview is on the record.

What is one identification worth?

BEFORE THE DAY — CANVASS

Who on Ferrier Street was actually asked in 2019

The file names two witnesses and the street had more people on it than that. Ask along the road until enough answers have come back to say whether nobody else saw anything or whether nobody else was asked.

PLAN CARD 114W

Wednesday. Three things are ready at once. Weill has worked out the pool of men who could have been on that street. Ilori has six hundred runs of the unit's own procedure, using arrays where the guilty man was not present. Baptiste wants one thing from you. She wants to know which finding goes at the top of a page the board will read in ten minutes. Today you work out what a witness pick multiplies. Then you find what the unit's own procedure cannot see. Last, you rank three claims by how hard each one is to argue with. The order they appear in is most of what the board will take away.

BRIEFING 15W

Three claims are ready and the board reads three weeks of work as one page.

WHAT YOU DECIDE 16W

Separate what a procedure can say about evidentiary weight from any unsupported numerical claim about guilt.

13.1 A multiplier still needs a starting point

TESTING & RECORDS • DATA ROOM • A ROOM • BALLPARK

SCENE 35W

Weill writes two deliberately hypothetical numbers on the board and refuses to call them Ward's odds. One is a starting odds assumption; the other describes how strongly an identification procedure discriminates in a test population.

GUIDE 50W

Five numbers, and three belong elsewhere: the multiplier for the other procedure, the size of the pool itself, and a resolution limit. Ask of each what this calculation needs. Evidence multiplies the odds you started with, so the answer depends on both assumptions. Neither of them can be left unstated.

ASK 23W

Under the stated hypothetical assumptions, calculate the updated odds—and then state why the case file is not entitled to call them Ward's odds.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

the exercise supplies a hypothetical prior rather than measuring one for Ward • a likelihood ratio multiplies prior odds; it does not create the starting odds

VERDICT 79W

Bayes' rule multiplies prior odds by a likelihood ratio. Here the exercise deliberately supplies prior odds of 1:4,000 and an illustrative likelihood ratio of 15, giving updated odds of about 1:267. That demonstrates base-rate dependence. It does not measure the prior probability that Ward was the offender, and a geographic "pool of men" is not a defensible substitute for such a prior. The review can use procedure performance to discuss evidentiary weight without manufacturing a case-specific probability of guilt.

TAKEAWAY 23W

A likelihood ratio says how evidence shifts odds; it does not supply the prior, so this exercise cannot become Ward's probability of guilt.

13.2 Six hundred arrays with no offender in them

IDENTIFICATION PROCEDURE · IDENTIFICATION SUITE · A ROOM · INJECT

SCENE 38W

Ilori has six hundred synthetic cases where the offender is known to be absent, and four versions of the procedure to push them through. What comes back is a measurement of the procedure rather than of anybody's guilt.

GUIDE 60W

Six hundred synthetic cases with the offender known to be absent, and four versions of the procedure. Run the ones you want and watch what comes back. Every pick in these runs is a false identification by construction, so what you are measuring is the procedure's own error rate rather than anybody's guilt. Recommend the version that actually buys something.

ASK 13W

Run the versions you want, then recommend the one that actually buys something.

BEHIND THE BUTTON 140W

Why a known-absent population is the instrument. If the offender is not in the array, a correct procedure produces no identification. Every pick is therefore an error, and counting them measures how often this procedure would accuse an innocent person — which cannot be measured from real cases, because nobody knows the truth in those.

What the versions differ in. The runs alter concrete system variables such as filler quality, the may-not-be-present instruction and whether the administrator knows the suspect. The combined reform changes several protections together, so it can test the package without isolating the contribution of every component.

Why 'buys something' rather than 'lowest error'. A version that eliminates false identifications by making witnesses pick nobody at all has bought nothing. The comparison has to hold correct identifications up against false ones, which is what the run reports.

TAKES AS READ

a procedure can be run on cases whose answer is already known · when the offender is absent, any pick is an error

VERDICT 86W

Every synthetic array here is offender-absent, so every identification is an error by construction. The useful question is where those errors land and how often the procedure produces them. A filler pick announces itself because the selected person is known not to be the suspect; a suspect pick can travel through a case as normal-looking evidence. The combined reform performs best in this authored test. Because several protections change together, the experiment supports that tested package rather than assigning the whole improvement to any one component.

TAKEAWAY 23W

Running a known population through your own procedure measures the procedure, and the errors it makes are the ones your real cases contain.

13.3 The order the board reads them in

INFLUENCE & THE ROOM · DELIBERATION ROOM · A PERSON · SEQUENCE

SCENE 38W

Three findings sit on one page and the board has ten minutes. Grace Ntuli places the cards along the corridor from the rebuilt corner to the interview room. She asks which finding should make the reader stop first.

GUIDE 73W

Order the three by how hard each is to argue with, hardest first. The rail is not a clock — all three findings are already established and go in the same report. What separates them is whether a reader can dispute the finding itself or only what follows from it, and a finding that rests on a measurement is in a different class from one that rests on a judgement about a person.

ASK 12W

Order the three findings by how hard each is to argue with.

BEHIND THE BUTTON 129W

What makes a finding hard to argue with. A quantity somebody can go and re-measure, with a stated method and an independent cross-check. The 34.2-metre sightline is that kind of finding. A missing line on the original 2019 form is nearly as direct.

What makes one more inferential. Administrator expectancy is a well-supported mechanism, but the 2019 file cannot measure how much it moved Cardoza. That is exactly why it belongs after the two findings that require less inference.

What makes one arguable. Anything resting on an inference about what a person knew, intended or transmitted. The expectancy finding is sound and it is the one a reader will contest, which is exactly why it belongs after the two that cannot be contested rather than in front of them.

TAKES AS READ

a reader gives most weight to what they read first

VERDICT 71W

A board reading quickly should meet the hardest-to-dispute findings first. The 34.2-metre geometry can be re-surveyed and independently checked against the lease plan. The missing caution is visible on the original 2019 form. The administrator's knowledge is also documented, but what that knowledge did in this particular case is an inference informed by controlled research. Putting the reproducible findings first makes the report's evidentiary hierarchy visible rather than relying on rhetoric.

TAKEAWAY 19W

Put the finding nobody can argue with first, because it is what the arguable ones are then read against.

13.4 What the review is looking for now

INFLUENCE & THE ROOM · DELIBERATION ROOM · A ROOM · SPOT

SCENE 42W

The file is larger than the review has time for. Ntuli sets one question the team is answering, and changes it as the picture develops — after the array form turns up, after the lux measurement, after the confidence statements are compared.

GUIDE 53W

Pull the documents the current question wants and leave the rest in the file. The question on the board changes during the day and nobody announces it. What is scored is the documents either side of a change, because they are the only ones that show whether the room is reading the board.

ASK 12W

Pull to the question on the board, and keep watching the board.

BEHIND THE BUTTON 92W

Why a review changes its question. It starts on procedure, because procedure is checkable. Once the light measurement lands it becomes what could have been seen. Once the two confidence statements are side by side it becomes what happened to the witness after the identification, which is a different set of documents entirely.

Why the changeover costs a review. Reading time is the only resource here, and a document read under the old question is read for the wrong thing — people do not re-read a document they have already ticked off.

TAKES AS READ

a review works to a stated question that somebody can change

VERDICT 136W

Three questions run across the day and each wants a different part of the file. Anything about how the procedure was run, then anything about what could have been seen, then anything about what the witness was told afterwards. A document can answer two at once, which is what makes the change cost real. The panel scores the window either side of each change, because most of the file is wanted by neither question and is correctly left unread by somebody who has read nothing. And a review is the worst case for a withdrawn instruction, because nobody re-reads a document they have already been through: read under the wrong question, it is not merely unhelpful — it is ticked, and the thing it would have shown is now behind a mark that says somebody looked.

TAKEAWAY 15W

The cost of a withdrawn instruction is paid by whoever is still working to it.

END OF THE DAY 16W

The strength of a piece of evidence and the strength of a case are different quantities.

The last reversible moment

PLAN CARD 110W

Thursday. This is the last day anything can still be added. Ines Baptiste has money for one more piece of work. She has four proposals in front of her. One of them is a fresh interview with Alma Cardoza. Ren Okada has spent two weeks explaining what that interview would cost. It is not money he is talking about. Samuel Oyelaran has been asked a harder question. Does Cardoza's distress about the case tell anybody anything at all? Today you choose what to buy. Then you read a test score against the group it was scored against. Last, you settle what a frightened witness's state does and does not prove.

BRIEFING 17W

After today the report is drafting, and one of the options on the table cannot be undone.

WHAT YOU DECIDE 11W

Decide what else to buy, and what asking again would cost.

14.1 Three days, and one irreversible purchase

MEMORY & INTERVIEW · INTERVIEW SUITE A · A ROOM · VALUE

SCENE 42W

Three days remain. Ren Okada lays four proposals on the last empty desk and turns the re-interview card face down. He says one purchase can create a new account without recovering the old one, while using time the unit cannot get back.

GUIDE 52W

Three days and four proposals. Open each and ask what its result would change about the review's conclusion. One of them creates a new memory report that cannot recover the pre-photograph state, while consuming time that could answer a still-open physical question. Weigh that against what it would tell you, then commit.

ASK 10W

Which work do you buy with the last three days?

BEHIND THE BUTTON 128W

Why a re-interview cannot recover the past. Asking again creates another retrieval episode, and the answer can be influenced by everything learned since 2019. The original 2019 document remains intact, but no 2026 interview can recreate the memory state that existed before later photographs, feedback and publicity. Scarce time should therefore go where a result can change the decision. be spent last.

What makes a purchase worth three days. Two possible results that point at different conclusions. A proposal that would confirm what the file already says has spent the unit's last window on reassurance.

Why the card is face down. Okada is making the irreversibility visible rather than letting it be discovered afterwards, which is the whole of what good evidence handling looks like from the inside.

TAKES AS READ

work that cannot change a conclusion is not evidence for it · some evidence is used up by being collected

VERDICT 84W

Evidence has value when it can change the recommendation. The independent survey answers a question the board has already asked and tests the unit's central physical finding. The camera record can settle another dated fact cheaply. More mock witnesses would narrow an interval that is already far from a fair six-person array. Re-interviewing Cardoza is different because it consumes something. It cannot recreate the pre-photograph memory state and may influence any later account, while the original 2019 document itself remains preserved as the baseline.

TAKEAWAY 15W

The best evidence purchase changes the decision without consuming information that cannot be recovered later.

14.2 One score, and what it is compared against

TESTING & RECORDS · DATA ROOM · A PERSON · BALLPARK

SCENE 36W

Alma Cardoza took a standard face-matching test for the unit in March and scored 42. Tomas Weill opens the age-band norms beside it: mean 51, standard deviation 7, with the rebuilt street visible through the glass.

GUIDE 52W

Five numbers, and two of them belong elsewhere: the highest score the test allows, and the surveyed distance. Ask of each whether this comparison needs it. A raw score means nothing without a distribution to sit in. And note the limits: the test was taken seven years later, with the pictures present.

ASK 14W

Estimate how far below the norm group's mean Cardoza's score falls, in standard deviations.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a raw score has no meaning without the distribution it came from · reliability: the same measurement twice, and two people measuring once — taken as read

VERDICT 77W

A raw score becomes interpretable only against a norm distribution. Cardoza's 42 is nine points below the mean of 51. Dividing that gap by the seven-point standard deviation gives 1.29 standard deviations below the mean. The signed z-score is therefore -1.29. That is low but not a diagnosis. The test was also taken seven years after the robbery. Face matching with pictures present is related to, but different from, identifying a remembered face from a dark street.

TAKEAWAY 24W

A standard score says how unusual a result is, which is the only form in which one person's score can be compared with anybody's.

14.3 What distress cannot tell you

STRESS, SLEEP & THE BODY · PHYSIOLOGY BAY · A ROOM · DEGENERACY

SCENE 32W

Oyelaran has been asked twice this week whether Cardoza's distress about the case supports her identification, and once whether it undermines it. He writes both questions on the board and answers neither.

GUIDE 53W

Two dials, one readout. Set the arousal at encoding and the arousal now, and watch the accuracy the model predicts. Find the settings that produce the accuracy actually observed — and then keep looking, because more than one pair produces it. The second measurement on the panel is what narrows the family down.

ASK 27W

Use the toy model to find multiple states that fit the same overall performance. What, if anything, does Cardoza's distress now establish about her identification in 2019?

BEHIND THE BUTTON 165W

Why the curve has two sides. Moderate arousal at encoding sharpens attention on central detail and improves what is remembered about it. High arousal narrows attention further and degrades the periphery, so accuracy for anything outside the centre falls. One accuracy figure therefore fits a low arousal and a high one.

What distress now is evidence of. Distress at interview is arousal at retrieval, and it is produced by having been in a frightening event, by being told the case is important, and by being asked repeatedly — which is why it cannot separate a witness who encoded well from one who did not. Both branches of the board's question predict it.

What does separate them. Central-detail accuracy against peripheral-detail accuracy, measured separately: high arousal at encoding predicts a specific pattern — good on the weapon, poor on the face — that moderate arousal does not. That is the second measurement, and it is why the answer is a family of possibilities until it arrives.

TAKES AS READ

distress can be assessed without assessing what caused it

VERDICT 96W

The inverted-U board is a teaching model of non-identifiability, not a physiological reconstruction of Cardoza. Different unobserved states can produce the same overall performance, and real stress-memory effects depend on timing, task and individual differences. Looking separately at central and peripheral details can test a prediction of an attention-narrowing model, but it still does not recover an exact arousal value from a seven-year-old record. Current distress is even less diagnostic because both a correct and an incorrect identification can be deeply distressing. The defensible conclusion is about limits on inference, not a diagnosis of the witness.

TAKEAWAY 13W

A psychological mechanism can suggest vulnerabilities without reconstructing an unmeasured state years later.

END OF THE DAY 19W

The value of a measurement is the decision it would change, and one of these cannot be taken back.

Sign the referral

PLAN CARD 107W

Friday. The report goes tonight. Ines Baptiste has the page in front of her. Ren Okada has a list of the things the review may not say. Marcus Ilori wants the recommendation to be about the procedure. He does not want it to be about Karen Loomis, who followed the policy of her year. Elias Ward has now served seven years and three weeks. Today you sort the review's claims by what stands behind each one. Then you settle what the recommendation actually asks the board to do. Last, you name which findings would still hold if every witness withdrew tomorrow. Nothing can be added after tonight.

BRIEFING 26W

Tonight you sign. The board is not deciding whether Ward is innocent; it is deciding whether this identification was strong enough to keep carrying his conviction.

WHAT YOU DECIDE 17W

Separate reproducible findings, witness-dependent findings and unresolved claims, then sign the recommendation the evidence can actually carry.

15.1 What stands behind each sentence

INFLUENCE & THE ROOM · DELIBERATION ROOM · A ROOM · CASEBOOK

SCENE 33W

Four sentences from the draft are on the table. Ntuli asks the room to put each one beside the kind of support it actually has, and warns that one of them has none.

GUIDE 57W

Four sentences from the draft, and four kinds of support. Pair them by asking who could check each one, and how. A survey can be repeated. An experiment can be rerun. A chronology sits in dated documents. And one of the four has no support here at all, which is why saying so protects the other three.

ASK 8W

Match each claim to what stands behind it.

BEHIND THE BUTTON 181W

Why explanations rather than labels. Naming a finding is not accounting for it, and a plausible-sounding mechanism attached to the wrong observation is the commonest way a wrong story survives. Committing an explanation to one clue means claiming it accounts for that clue specifically and not for its neighbour, which is where the two come apart.

How to use the one-each rule. The explanations are a set to be distributed, not a list to be sampled, so every join constrains the rest. Settling the two you are confident of can decide the remaining pair by elimination. Where it does not, two explanations are still competing for one clue, and that competition is the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every explanation used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

claims of different kinds cannot be defended in the same way

VERDICT 72W

Different claims require different kinds of support. A distance can be checked by another survey. A lineup effect can be tested by repeating an experiment. A chronology can be checked against dated records. The claim that Ward was not the man has none of those supports in this review. Leaving that sentence out protects the others. A report gains credibility when each claim has matching evidence. Unsupported conclusions should remain explicitly unresolved.

TAKEAWAY 25W

Every sentence in a report is answerable in a particular way, and the ones that are not answerable at all should not be in it.

15.2 What holds if everybody withdraws

PERCEPTION & THE SCENE · RECONSTRUCTION BAY · A PERSON · CHOICE

SCENE 34W

Vision scientist Frey shuts the bay lamps down for the last time. She repeats the question from the first morning. Which findings would remain true if nobody involved in the case ever spoke again?

GUIDE 53W

Four findings, and the question is which survives everybody withdrawing. Ask of each whether it rests on a memory, a form, or a measurement. A distance can be remeasured. A utility record and a reconstruction can be repeated. The others still matter and depend on accounts, which is exactly what the question removes.

ASK 13W

Which evidence would remain available if every witness in the case withdrew tomorrow?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a claim about a place can be checked without the people who were there

VERDICT 83W

Some evidence survives because it is not stored in a person's memory. The sightline can be re-surveyed, the utility outage is a dated external record, and the reconstruction method can be repeated under stated conditions. Those are different kinds of support—a measurement, a historical record and a modelled experiment—but none depends on asking Cardoza or Aktaş to remember again. The confidence statement and co-witness agreement remain important records, but they are records of people's accounts rather than independently reproducible properties of the scene.

TAKEAWAY 25W

The strongest spine of a contested review separates repeatable measurements, dated external records and witness-dependent accounts instead of treating them as one kind of evidence.

15.3 Put your name on it

IDENTIFICATION PROCEDURE • IDENTIFICATION SUITE • A ROOM • CHOICE

SCENE 37W

Ines Baptiste has three versions of the closing sentence and has struck out two. Marcus Ilori reads the survivor at the end of the sightline. The board packet closes tonight, after seven years of a fourteen-year sentence.

GUIDE 50W

Four recommendations, and the review has weighed one piece of evidence without reconstructing the robbery. Ask of each whether it claims more than that. One asserts innocence. One blames an individual for following the policy then in force. One waits for evidence nobody is looking for. The packet closes tonight.

ASK 10W

What do you sign and send to the board tonight?

BEHIND THE BUTTON 195W

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TAKES AS READ

a review recommends a course of action rather than reaching a verdict • storage and forgetting: the curve is steep at the start — taken as read

VERDICT 80W

The review can assess the weight of the identification without reconstructing the robbery itself. Referral asks the body with authority to reopen a conviction to examine evidence that cannot carry the role assigned to it. The same procedural finding also matters beyond Ward because Loomis followed the policy then in force. Discipline would misplace the failure on an individual. An innocence finding would go further than the evidence. The recommendation should match the strongest claim the review can actually support.

TAKEAWAY 17W

A recommendation should be no broader than the evidence chain that can still support it under challenge.

END OF THE DAY 18W

A review that says only what it can support is the only kind a board can act on.

THE CLOSING CARD

The board chair signs the referral before the room empties. The board accepts the thirty-four-metre survey, the utility record showing column 4471 was out, the mock-witness evidence that the array pointed toward Ward, and the dating of Cardoza's confidence. It does not decide who stood in the doorway. Eleven months later the appeal court sets the conviction aside because the identification cannot carry the conviction standing alone. Ward leaves Hallam Correctional after eight years and one month, and the court says explicitly that it is not finding him innocent.

What it cost: Alma Cardoza learns from a newspaper that the identification she gave in good faith has been dissected without another interview, because the unit chose not to create a new 2026 account and mistake it for the lost 2019 memory state. Karen Loomis is named in the judgment for a procedure that was policy when she ran it. What is unfinished: Hallam used the same policy until 2021, nobody has counted all of those arrays, and the unit receives enough funding to review only the cases in which somebody is still incarcerated.

You did not prove Ward innocent and you did not need to. You measured the sightline instead of arguing about eyesight, separated a correct lighting calculation from a false premise, traced later details to later sources, and refused to turn a psychological tendency into a certainty about one witness. Then you signed only what the evidence could carry. A man walked out after eight years because the review finally asked the right question.

Card text only — no options, boards or answers. 11,585 words of card prose across 53 stops. Generated from `themes/sightline` by `tools/make-cardbook.mjs`.